Object-Oriented Analysis and Design

CA1-19-LPCH Kerry Local Food Hub

https://github.com/Geese4000/ca-analysis.git

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1. Scenario Interpretation

1.1 Organisation and current process

Kerry Local Food Hub collects vegetables, fruit, eggs, preserves and other products from small producers and prepares mixed orders for households, cafés, accommodation providers and community programmes. Producers notify the hub of expected deliveries; staff and volunteers check goods on arrival, store them, allocate stock to customer orders, pack and label orders, and hand them over to customers or drivers.

All of this is currently coordinated through spreadsheets, messages and paper checklists (E-01). Decisions about shortages, substitutions and priorities are made informally, case by case (E-08, E-09).

1.2 Problem statement

The hub cannot reliably tell how much of each product it actually has available to sell. A quantity moves through several states (expected, received, held, accepted, allocated), but these states are not recorded separately, so staff treat goods as available before checks are complete and must reallocate at short notice when deliveries differ from what was expected (E-02, E-03, E-04). Because customer substitution preferences and shortage priorities are not recorded, allocation decisions are inconsistent, and the link between an allocated item and its producer and batch is lost once stock is split across orders (E-05, E-09, E-07). The consequences are orders reopened after packing, problems discovered only at handover, avoidable waste of perishables, customers receiving substitutes they did not want, and weak traceability (E-10).

1.3 Desired outcomes

- Staff know, at any point, how much of each product is expected, received, held, accepted and available.
- Shortages are handled consistently, using recorded customer preferences and agreed priorities.
- Customers who accept reduced quantities are told before delivery, not at handover.
- Any allocated item can be traced back to its producer and batch.
- The hub manager can see deliveries, held stock, shortages, substitutions and producer contributions without compiling them by hand.
- Volunteers can do receipt and packing work without seeing commercial terms.

1.4 Evidence and uncertainty

ID	Evidence	Source	Type
E-01	Deliveries, stock, orders and collections are managed through spreadsheets,	Scenario: Organisation and current service	Stated

ID	Evidence	Source	Type
	messages and paper checklists		
E-02	“Expected quantity and accepted quantity are often different.”	Hub manager	Stated
E-03	“The crop may be ready, but the exact quantity is not known until harvest.”	Producer	Stated
E-04	“Received does not mean available if quality checking is incomplete.”	Packing volunteer	Stated
E-05	“I will accept some substitutions, but not every alternative.”	Customer	Stated
E-06	“A smaller quantity may be useful if I know before delivery.”	Hospitality customer	Stated
E-07	“We need to know which producer and batch supplied an item after it has been allocated.”	Traceability coordinator	Stated
E-08	Products are supplied individually, by weight, box or case; staff convert units manually	Scenario: Producers, products and expected deliveries	Stated
E-09	Shortage priority between customer groups is informal; preferences are not always recorded	Scenario: Orders and allocation	Stated
E-10	Late deliveries can force orders to be reopened after packing; problems are found at handover	Scenario: Orders and allocation; Collection	Stated

2. Purpose, Scope and Boundary

2.1 Purpose

To give the hub a single, reliable record of products as they move from expected delivery to customer handover, so that staff can allocate available stock fairly, keep customers informed of shortfalls, and trace every allocated item to its produce and batch.

2.2 In scope

- Recording expected deliveries from producers
- Recording receipt and quality decisions (accept, partially accept, hold, reject) per delivery line
- Tracking stock state: recieved, held, accepted, available, allocated
- Recording customer orders and their substitution and reduced-quantity preferences
- Allocating available stock to order lines, including during shortages
- Flagging customers who need to be told about a shortfall before delivery
- Packing, labelling and reopening orders
- Recording handover exceptions (missing, damaged, wrong items) and returns
- Recording traceability details at receipt and keeping them linked through allocation
- Manager reporting on deliveries, held stock, shortages, substitutions, prepared orders and producer contributions
- Restricting what volunters can see

2.3 Out of scope

- Detailed payment and invoicing (stated in scenario); only the *fact* that a price adjustment was agreed is recorded (A-07)
- Route optimisation for deliveries (stated in scenario); only the handover to a driver is recorded
- Food-safety certification systems (stated in scenario)
- Storage space and capacity planning
- Producer and customer self-service portals (A-01, A-06)

2.4 Dependencies and constraints

- **Dependency:** producers notify the hub of expected deliveries before arrival.
- **Dependency:** physical quality checks are done by staff; the system records the outcome but does not perform the check.
- **Constraint:** receipt and packing are largely done by volunteers, who may have limited training and time.
- **Constraint:** volunteers must not see commercial terms.

- **Constraint:** perishable goods have short shelf lives, so decisions on held and returned goods are time-sensitive.
- **Constraint:** small occasional producers supply inconsistent records.

3. Uncertainty Register

3.1 Assumptions

ID	Assumption	Stands in for	Rationale
A-01	Staff record expected deliveries from producer messages; producers do not use the system directly	Q-01	Scenario says producers “notify” the hub; simplest reversible option
A-02	Each product has one selling unit; staff record the supplied unit and quantity at receipt alongside the converted selling quantity	Q-02	Keeps the original record while giving one comparable figure for availability
A-03	Producer and delivery date are mandatory at receipt; batch and use-by are recorded when supplied	Q-03	Small producers’ records are inconsistent (scenario), so stricter rules would block receipt
A-04	Any receiving staff member, including volunteers, may accept or hold goods; only the hub manager may downgrade or reject	Q-04	Rejection affects producers and pricing, so it is treated as a manager decision
A-05	During a shortage, stock is allocated by order date until priority rules are agreed	Q-05	Neutral, transparent and easy to replace
A-06	Customers state allowed substitutions and reduced	Q-06	Matches the customer quote: some substitutions, not all

ID	Assumption	Stands in for	Rationale
	quantities when ordering; staff record them; anything else requires contacting the customer		
A-07	An agreed price adjustment is recorded as a note on the delivery line, with no amount	Q-08	Keeps payment out of scope while preserving the decision
A-08	Held goods become available only after a completed quality check; returned perishables never re-enter stock	Q-07	Supported by the volunteer quote; cautious for food
A-09	Volunteers see product, quantity, order reference and customer name, but no prices or commercial terms	Q-09	Volunteers need customer names to label and hand over orders
A-10	Customers do not use the system; staff record orders and preferences received by message, phone or in person	Q-12	Scenario describes no customer self-service; keeps scope realistic
A-11	Thresholds in the non-functional requirements (steps, seconds, product count) are working targets, not agreed figures	Q-13	Lets NFRs be measurable before the hub confirms its expectations

3.2 Clarification questions

ID	Question	Ask	Why it matters	Blocks
Q-01	Do producers enter expected deliveries themselves, or do	Hub manage r,	Decides whether Producer is an actor or an offstage stakeholder	Actor-goal table, UC-01

ID	Question	Ask	Why it matters	Blocks
	staff record them?	Producer		
Q-02	How should products supplied by box or case be converted to selling units, and is the conversion fixed per product?	Hub manager	Availability figures are meaningless if units are inconsistent	FR-04, Product `sellingUnit`
Q-03	Which traceability details are mandatory at receipt, and what happens if a small producer cannot supply them?	Traceability coordinator	Determines whether incomplete deliveries are held or accepted	FR-05, A-03
Q-04	Who may accept, downgrade or reject delivered goods, and must the producer be told?	Hub manager	Defines the receiving actor and a permission rule	UC-02, UC-03
Q-05	When stock is short, how should hospitality customers, pre-paid boxes and community programmes be prioritised, and who decides?	Hub manager	Allocation cannot be specified fairly without it	FR-09, FR-10, A-05
Q-06	How do customers record acceptable substitutions and reduced quantities, and must each substitution be approved?	Customer, Hub manager	Prevents unwanted substitutes at handover	FR-08, BR-05, GH-04
Q-07	When may held or returned goods	Hub manage	Directly affects the availability figure	FR-06, FR-11, BR-04

ID	Question	Ask	Why it matters	Blocks
	return to available stock, and who decides?	r		
Q-08	Should agreed price adjustments be recorded, and with what detail, given payment is out of scope?	Hub manage r	Sets the scope boundary for commercial data	Scope, A-07
Q-09	Which producer, customer and commercial details may volunteers see?	Hub manage r	Defines the access-control NFR and rule	NFR-01, BR-02, A-09
Q-10	What does “producer contributions” mean in the manager’s reports: quantity supplied, accepted or sold?	Hub manage r	Changes what must be recorded per producer	FR-13
Q-11	Who decides to reopen a packed order when a late delivery arrives, and must the customer be told?	Hub manage r	Defines an alternative flow and possible rule	UC-07 Pack order
Q-12	Do customers place orders and state preferences directly, or through staff?	Hub manage r, Custom er	Decides whether Customer is an actor	Actor-goal table, UC-04
Q-13	How quickly must volunteers be able to record a receipt, and how many products does the hub handle at peak?	Hub manage r, Packing volunte er	Sets realistic NFR thresholds	NFR-02, NFR-03

3.3 Risks

ID	Risk	Impact	Mitigation
R-01	Priority rules (Q-05) stay unresolved	Allocation requirements may need rework	Keep allocation logic behind A-05 and flag dependent requirements
R-02	Volunteers skip recording steps under time pressure	Availability and traceability data become unreliable	Usability NFR limiting steps per task
R-03	Small producers cannot supply batch information	Traceability gaps for some items	A-03 allows receipt without batch, recorded as missing
R-04	Unit conversion rules differ between products	Incorrect availability figures	Q-02 raised early; record supplied and converted quantities (A-02)

4. Stakeholders and Actors

The hub sits between two groups it does not control: producers, whose quantities are uncertain until harvest (E-03), and customers, whose expectations differ by type (Orders section). The people who actually record information are inside the hub: the manager, staff and volunteers. The traceability coordinator is a direct stakeholder with a narrow but strict need: to answer “where did this come from?” after stock has been split (E-07). Influence is highest for the hub manager, who sets priorities and owns the open policy questions, so most clarification questions (Section 3.2) are addressed to them. Volunteers have less formal influence but carry out the most frequent tasks, which is why usability (NFR-02) and visibility limits (NFR-01) matter.

4.1 Stakeholder catalogue

Type: **Direct** = uses the system or is affected by it day to day; **Indirect** = affected by its outcomes; **External** = outside the hub organisation.

ID	Stakeholder	Type	Goals	Concerns	Influence	Role	Evidence
SH-01	Hub manager	Direct	Know what is expected, receive	Reallocation at short notice; informal priorities;	High	Primary actor	E-02; Collection and reporting section

ID	Stakeholder	Type	Goals	Concerns	Influence	Role	Evidence
			d, held and available; run allocation fairly; report on producers and shortages	manual reporting			
SH-02	Hub staff member	Direct	Record deliveries and orders accurately; allocate stock; resolve shortages and handover problems	Unclear availability; unrecorded preferences	Medium	Primary actor	Receipt; Orders and allocation sections
SH-03	Hub volunteer (receipt and packing)	Direct	Record receipt quickly; pack the right goods for each order	Treating unchecked goods as available; time pressure	Medium	Primary actor	E-04; volunteers process receipt and packing
SH-04	Traceability coordinator	Direct	Find the produc	Batch link lost after allocation;	Medium	Primary actor	E-07

ID	Stakeholder	Type	Goals	Concerns	Influence	Role	Evidence
			er and batch behind any allocated item	inconsistent producer records			
SH-05	Producer	External	Have produce accepted and paid for fairly; know outcomes of deliveries	Quantities uncertain until harvest; rejections and downgrades	Medium	Offstage stakeholder (A-01)	E-03
SH-06	Household customer (incl. pre-paid boxes)	External	Receive the ordered box, or acceptable substitutes	Unwanted substitutes; missing items	Low	Offstage stakeholder (A-10)	E-05
SH-07	Hospitality customer (cafés, accommodation)	External	Receive reliable quantities, or early notice of shortfalls	Late discovery of shortages	Medium	Offstage stakeholder (A-10)	E-06
SH-08	Community programme	External	Receive allocated	Being deprioritised under informal	Low	Offstage stakeholde	Orders and allocation section

ID	Stakeholder	Type	Goals	Concerns	Influence	Role	Evidence
			produce for their participants	rules		r	
SH-09	Driver	Direct	Collect the correct, complete orders	Discovering problems at handover	Low	Offstage stakeholder	Collection section
SH-10	Food-safety regulator	Indirect	Hub can show where food came from	Gaps in traceability records	Medium	Offstage stakeholder	Implied by E-07; certification itself is out of scope

4.2 Actor rationale

- **Actors are the people who record or query information in the system.** The hub manager, hub staff members, hub volunteers and the traceability coordinator all do this.
- **Generalisation:** a hub staff member can do everything a volunteer can, and a hub manager can do everything a staff member can. This keeps the use-case diagram readable and reflects the volunteer visibility limit (BR-02).
- **Producers and customers are offstage stakeholders** under A-01 and A-10, because staff record their deliveries and orders. If Q-01 or Q-12 is answered differently, they become primary actors.
- **The driver is offstage.** Staff record the handover; the driver does not interact with the system.
- **Offstage concerns still drive requirements:** customer concerns drive FR-08 and FR-10; producer concerns drive FR-01 and FR-13; the regulator's concern supports FR-12 and NFR-05.

4.3 Actor-goal table

Actor	Type	Goal	Use case	Requirements
Hub staff member	Primary	Record what a	UC-01 Record expected delivery	FR-01

Actor	Type	Goal	Use case	Requirements
		producer expects to deliver		
Hub volunteer	Primary	Record a delivery and its quality outcome	UC-02 Receive delivery	FR-02, FR-03, FR-04, FR-06
Hub volunteer	Primary	Record the producer and batch of received goods	UC-11 Record traceability details	FR-05
Hub staff member	Primary	Release held goods once checked	UC-03 Release held stock	FR-06
Hub staff member	Primary	Record a customer order and its preferences	UC-04 Record customer order	FR-07, FR-08
Hub staff member	Primary	Allocate available stock to orders	UC-05 Allocate stock to orders	FR-09
Hub staff member	Primary	Resolve a shortfall using customer preferences	UC-06 Handle shortfall	FR-08, FR-10
Hub volunteer	Primary	Pack and label an order	UC-07 Pack order	FR-09
Hub staff	Primary	Record handover	UC-08 Record	FR-11

Actor	Type	Goal	Use case	Requirements
member		problems and returns	handover	
Traceability coordinator	Primary	Find the producer and batch of an allocated item	UC-09 Trace item	FR-12
Hub manager	Primary	Review deliveries, stock, shortages and producer contributions	UC-10 View hub summary	FR-13

5. Requirements Catalogue

Requirements were derived by following each quantity through the hub, from expected to received, held or accepted, allocated and handed over, and asking what the system must record at each step to answer the three questions in the problem statement: what is available, who gets it, and where it came from. Each requirement states one capability, cites its evidence and avoids design detail such as screens or storage.

The Must set is deliberately limited to what closes those three questions. Unit conversion (FR-04) is important but rated Should because the conversion policy is unconfirmed (Q-02); specifying it as Must would commit the hub to a rule nobody has agreed. Reporting (FR-13) depends entirely on data captured by the Must requirements, so it can follow in a later release without losing information.

Priorities use MoSCoW. **Must** = without it the core problem (unreliable availability, unfair allocation, lost traceability) is not solved. **Should** = important but the hub could operate for a first release without it. **Could** = useful if time allows. Verification refers to Gherkin scenarios (GH-xx), inspection (INS) or test (TST).

5.1 Functional requirements

ID	Requirement	Source	Priority	Rationale for priority	Verification
FR-	The system shall record each	E-03; manager	Must	Expected vs accepted comparison is the starting	TST

ID	Requirement	Source	Priority	Rationale for priority	Verification
01	expected delivery line with producer, product, expected quantity and expected date.	reporting needs		point of the problem (E-02)	
FR-02	The system shall record the received quantity of each delivery line separately from its expected quantity.	E-02	Must	Core cause of reallocation errors	GH-01
FR-03	The system shall record one receipt outcome per delivery line (accepted, partially accepted, held or rejected) together with the accepted quantity.	Receipt section	Must	Outcomes differ within one delivery	GH-01, GH-02
FR-04	The system shall record the supplied unit and quantity of a delivery line alongside its quantity in the product's selling unit.	E-08; A-02	Should	Needed for accurate availability, but conversion rules are unconfirmed (Q-02)	TST

ID	Requirement	Source	Priority	Rationale for priority	Verification
FR-05	The system shall record the producer, batch code and, where supplied, harvest or use-by date for each received delivery line, and shall record when batch information is missing.	E-07; A-03	Must	Traceability starts at receipt	GH-05
FR-06	The system shall exclude held and uninspected quantities from available stock until a quality check releases them.	E-04; BR-01	Must	“Received does not mean available” is the central rule	GH-02
FR-07	The system shall record a customer order as one or more order lines, each with a product and requested quantity.	Orders section	Must	Allocation needs order lines	TST
FR-08	The system shall record, for each order line, whether the customer accepts a	E-05; E-06; A-06	Must	Prevents unwanted substitutes; preferences are currently lost	GH-03, GH-04

ID	Requirement	Source	Priority	Rationale for priority	Verification
	reduced quantity, removal of the item, and which substitute products are permitted.				
FR-09	The system shall allocate a quantity from a specific stock lot to a specific order line.	E-07; allocation section	Must	Allocation is the link that makes traceability possible	GH-03, GH-05
FR-10	The system shall identify order lines that cannot be fully allocated and flag customers who must be contacted before delivery.	E-06	Should	High value to hospitality customers; can initially be handled by phone	GH-03
FR-11	The system shall record handover exceptions (missing, damaged, incorrectly allocated) and the stock decision for any returned goods.	Collection section; BR-04	Should	Problems at handover are frequent but secondary to availability	TST
FR-	The system shall identify	E-07	Must	Directly requested by the	GH-05

ID	Requirement	Source	Priority	Rationale for priority	Verification
12	the producer and batch of any allocated quantity on an order.			traceability coordinator	
FR-13	The system shall provide the hub manager with a summary of expected vs accepted quantities per producer, held stock, available stock, shortfalls, substitutions and prepared orders.	Manager information needs	Should	Replaces manual reporting; depends on the data captured by Must items	INS

Won't have (this release): payment and invoicing, route optimisation, food-safety certification, storage capacity planning, producer and customer self-service (A-01, A-10).

5.2 Non-functional requirements

ID	Category	Requirement	Source	Priority	Verification
NFR-01	Security / access control	A user in the volunteer role shall be able to view 0 price, payment or commercial-term fields.	Scenario (visibility); BR-02	Must	TST: log in as volunteer and inspect every screen and report for commercial fields
NFR-02	Usability	A volunteer who has had a 15-minute briefing shall be able to record the receipt and outcome of one delivery line in under 60	Volunteers under time pressure; A-11	Should	TST: timed task with 3 volunteers

ID	Category	Requirement	Source	Priority	Verification
		seconds.			
NFR-03	Performance	The available quantity of a product shall be displayed within 2 seconds for 95% of requests with up to 500 active products.	Rapid reallocation (E-01); A-11	Should	TST: load test
NFR-04	Data integrity	The available quantity of a stock lot shall never exceed its accepted quantity minus allocated and written-off quantities.	E-04; BR-01	Must	TST: automated check after every allocation and release
NFR-05	Auditability	100% of receipt outcomes, quality releases and allocations shall record who made them and when.	E-07; regulator concern	Must	INS: sample 20 records

All thresholds are working targets (A-11) pending Q-13.

6. Business Rules and Constraints

A business rule here is a policy the hub follows regardless of any system. Only statements supported by the scenario or a stakeholder quote are listed as rules. BR-05 is marked *Derived* because the customer quote supports the principle that not every substitute is acceptable, but the approval process behind it is still open.

6.1 Business rules

ID	Rule	Evidence	Confidence	Enforced by
BR-01	Goods that are held or awaiting quality checking	E-04	Stated	FR-06, NFR-04

ID	Rule	Evidence	Confidence	Enforced by
	are not available for allocation.			
BR-02	Volunteers may not see commercial terms.	Scenario: volunteers “without seeing all commercial terms”	Stated	NFR-01
BR-03	Every allocated quantity must remain linked to the producer and batch it came from.	E-07	Stated	FR-09, FR-12
BR-04	Returned perishable goods do not automatically return to available stock; a decision must be recorded.	Collection section	Stated	FR-11
BR-05	A substitute product may only be supplied if the customer has	E-05	Derived from stakeholder statement; approval process unconfirmed (Q-06)	FR-08

ID	Rule	Evidence	Confidence	Enforced by
	permitted it or approved it.			

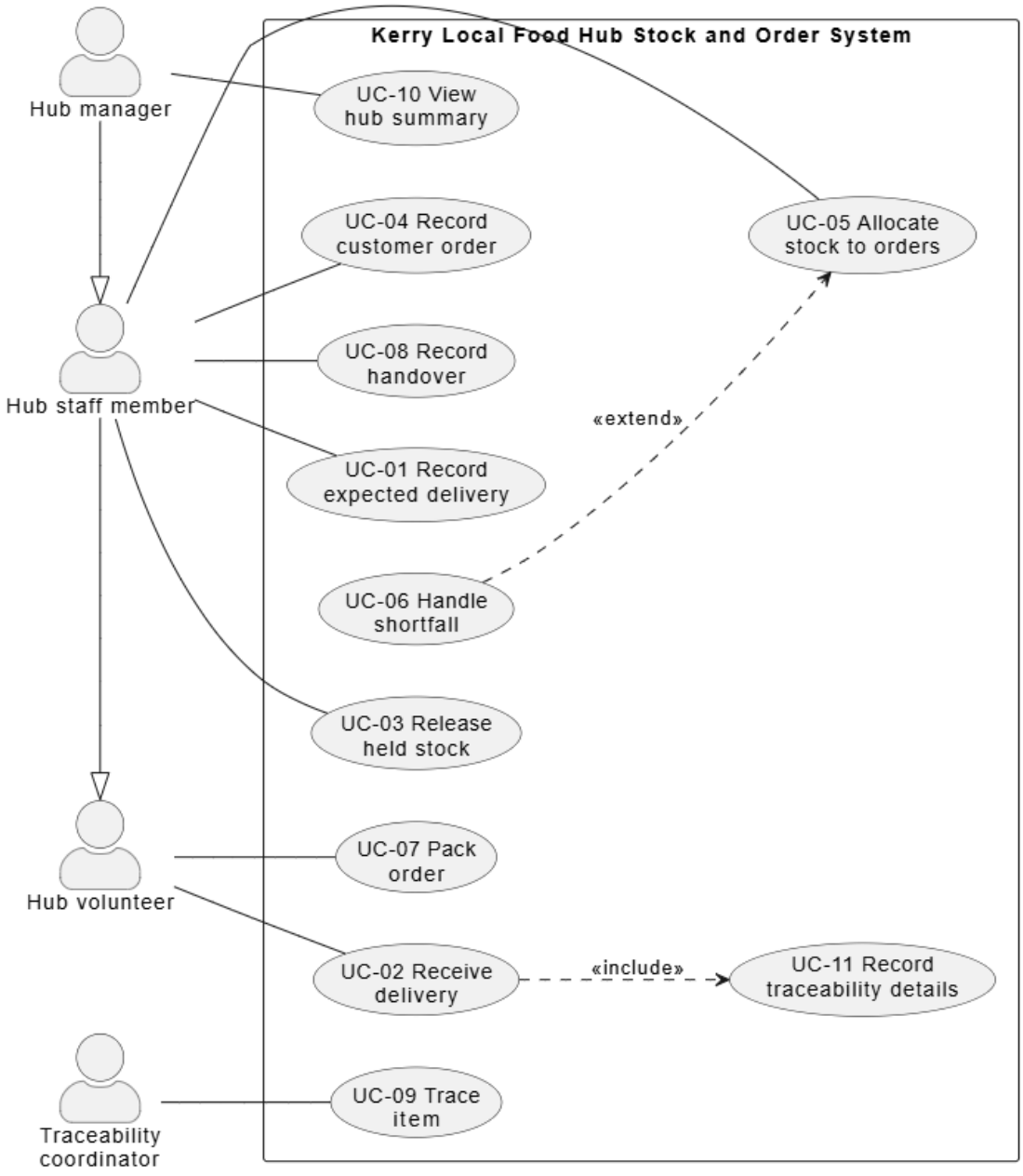
Not rules yet: shortage priority (Q-05, A-05), authority to downgrade or reject (Q-04, A-04), mandatory traceability fields (Q-03, A-03) and approval of reduced quantities (Q-06, A-06) remain assumptions until stakeholders confirm them.

6.2 Constraints

- CON-01: Receipt and packing are performed mostly by volunteers with limited training time.
- CON-02: Perishable goods must be handled within short time windows.
- CON-03: Producer records are inconsistent, particularly for small occasional suppliers.
- CON-04: Payment, route optimisation and certification are excluded by the hub.

7. Use-Case Model

7.1 Use-case diagram



Use-case model

Source: `diagrams\use-case-model.puml`. The system boundary is **Kerry Local Food Hub Stock and Order System**. Payment, routing and certification sit outside the boundary (Section 2.3).

7.2 Use-case catalogue

ID	Use case	Primary actor	Goal	Requirements
UC-01	Record expected delivery	Hub staff member	Note what a producer expects to deliver and when	FR-01
UC-02	Receive delivery	Hub volunteer	Record what arrived and the outcome for each line	FR-02, FR-03, FR-04, FR-06
UC-03	Release held stock	Hub staff member	Make checked goods available, or reject them	FR-06
UC-04	Record customer order	Hub staff member	Capture order lines and substitution preferences	FR-07, FR-08
UC-05	Allocate stock to orders	Hub staff member	Assign available stock lots to order lines	FR-09
UC-06	Handle shortfall	Hub staff member	Apply customer preferences when	FR-08, FR-10

ID	Use case	Primary actor	Goal	Requirements
			stock is short	
UC-07	Pack order	Hub volunteer	Assemble and label an order from its allocations	FR-09
UC-08	Record handover	Hub staff member	Record collection, exceptions and returns	FR-11
UC-09	Trace item	Traceability coordinator	Find the producer and batch of an allocated item	FR-12
UC-10	View hub summary	Hub manager	Review deliveries, stock, shortfalls and producer contributions	FR-13
UC-11	Record traceability details	Hub volunteer	Capture producer, batch and dates at receipt	FR-05

Relationships

- UC-02 Receive delivery **«include»** UC-11 Record traceability details: every receipt records traceability (BR-03).

- UC-06 Handle shortfall **«extend»** UC-05 Allocate stock to orders: it runs only when available stock is less than the quantity ordered.
- Actor generalisation: Hub manager → Hub staff member → Hub volunteer (Section 4.2).

Model rationale. Each use case is one actor goal from the actor-goal table, named as a verb and an object; there are no "manage" use cases that hide several goals. Receive delivery is split from Release held stock because they happen at different times and often by different people: a volunteer records arrival, while a staff member later completes the check. «include» is used once, because recording traceability is a mandatory part of every receipt. «extend» is used once, because shortfall handling only happens when supply is short. Reopening a packed order is treated as an alternative flow of UC-07 rather than a separate use case, since who may reopen orders is still unknown (Q-11).

7.3 Detailed use case: UC-02 Receive delivery

UC-02 was chosen for detailed description because it contains the scenario's richest variation (accept, partially accept, hold and reject outcomes) and is where availability and traceability begin.

Field	Content
Primary actor	Hub volunteer
Stakeholders and interests	Hub manager: accurate accepted quantities (E-02). Producer: fair record of what was delivered (E-03). Traceability coordinator: batch recorded (E-07). Customers: only checked goods are allocated (E-04).
Trigger	A producer's delivery arrives at the hub.
Preconditions	The volunteer is identified in the system. The producer exists in the system.
Success guarantee (postconditions)	Each delivery line has a received quantity, an outcome and traceability details. Accepted quantities are available stock; held quantities are not available (BR-01). Who recorded the receipt and when is stored (NFR-05).

Main success scenario

1. The volunteer identifies the producer and selects the matching expected delivery.
2. The system lists the expected delivery lines with products and expected quantities.
3. For each line, the volunteer records the received quantity and supplied unit.
4. The system shows the received quantity in the product's selling unit (FR-04).
5. The volunteer records the traceability details (include UC-11).

6. The volunteer records the outcome as accepted.
7. The system records the accepted quantity as available stock in a new stock lot.
8. Steps 3–7 repeat for each line.
9. The volunteer confirms the delivery is complete; the system records the receipt with user and time.

Alternative flows

- **1a. No expected delivery exists** (unannounced delivery): the volunteer creates the delivery lines during receipt; the system marks the delivery as unannounced.
- **3a. Received quantity differs from expected:** the system records both quantities and highlights the difference for the manager summary (FR-13).
- **6a. Only part of the line is acceptable:** the volunteer records "partially accepted" and the accepted quantity; only that quantity becomes available. The remainder is recorded as held or rejected.
- **6b. Quality check cannot be completed now:** the volunteer records "held" with a reason; the quantity is stored but not available until UC-03 releases it (BR-01).

Exceptions

- **5a. Batch information missing:** the volunteer records that batch is missing (A-03); receipt continues and the line is marked for follow-up.
- **6c. Goods appear unacceptable:** under A-04 a volunteer cannot reject goods; the volunteer records "held" and the line is referred to the hub manager, who may record "rejected" via UC-03.
- **Any step. The volunteer abandons receipt:** lines already recorded are saved as held so that no unchecked stock becomes available.

Open issues: Q-02 (unit conversion), Q-03 (mandatory traceability fields), Q-04 (rejection authority).

8. Behavioural Examples (Gherkin)

The five scenarios were chosen to exercise the rules with the strongest evidence rather than the easiest flows. GH-01 and GH-02 test the difference between received, accepted and available stock, which is the core of the problem statement. GH-03 is a boundary case: the order is short by exactly 1 kg and the customer accepts a reduced quantity, so the outcome depends on a recorded preference. GH-04 is a failure case showing that an unpermitted substitute is refused and stock is left untouched. GH-05 shows traceability surviving a split across two batches. All steps use domain language and concrete values; none mention screens, buttons or storage. GH-03 depends on assumption A-05, which is stated in the feature file.

Feature files are in `features/`. Each scenario is tagged with its requirement, rule and use case.

ID	Scenario	Kind	File	Links
GH-01	Partially accept a delivery line	Success / alternative	`receive_delivery.feature`	FR-02, FR-03, UC-02
GH-02	Held stock is not available for allocation	Rule / boundary	`receive_delivery.feature`	FR-06, BR-01, UC-02, UC-03
GH-03	Shortfall of a line where reduced quantity is accepted	Boundary / alternative	`allocate_stock.feature`	FR-08, FR-10, UC-06
GH-04	Refuse a substitute the customer has not permitted	Failure / rule	`allocate_stock.feature`	FR-08, BR-05, UC-06
GH-05	Trace an allocated item to producer and batch	Success	`trace_item.feature`	FR-05, FR-09, FR-12, BR-03, UC-09

9. Domain Analysis

9.1 Candidate-concept analysis

Candidates were extracted from every noun and noun phrase in the scenario and stakeholder quotes, then classified. Decisions: **Keep** (becomes a class), **Merge** (synonym), **Convert** (attribute, association, status or enumeration), **Remove** (out of scope, technology or too vague).

ID	Candidate	Source	Classification	Decision	Rationale
C-01	Producer	Throughout	Conceptual class	Keep	Supplies deliveries; needed for traceability and contribution reports
C-02	Small occasional supplier	Producers section	Synonym	Merge → Producer	Same concept; inconsistency is handled by optional batch attributes
C-03	Product	Throughout	Conceptual class	Keep	Requested, delivered and

ID	Candidate	Source	Classification	Decision	Rationale
					substituted
C-04	Variety, size, production method, packaging	Producers section	Attribute	Convert → Product attributes	Distinguish similar products; no identity of their own (`variety` kept; others deferred)
C-05	Unit (weight, box, case, individual)	Producers section	Attribute	Convert → `sellingUnit`, `suppliedUnit`	Conversion rules unknown (Q-02); a separate class would invent policy
C-06	Expected delivery	Producers section	Synonym	Merge → Delivery	An expected delivery becomes a received one; modelled as one Delivery with dates
C-07	Delivery	Receipt section	Conceptual class	Keep	Groups lines from one producer arrival
C-08	Delivery line	Implied by “partially accepted”	Conceptual class	Keep	Outcomes differ per product within one delivery (E-02)
C-09	Expected / received / accepted quantity	E-02	Attribute	Convert → DeliveryLine attributes	Values of one line at different stages
C-10	Available quantity	E-04	Attribute (derived)	Convert → `/availableQuantity` on StockLot	Calculated from accepted, allocated and written-off quantities

ID	Candidate	Source	Classification	Decision	Rationale
					(NFR-04)
C-11	Accepted / partially accepted / held / rejected	Receipt section	Status	Convert → ReceiptOutcome enumeration	States of a delivery line
C-12	Quality check / inspection	E-04; Receipt section	Conceptual class	Keep as QualityCheck	Has its own time, result and person; a line may be checked more than once
C-13	Grade	Receipt section	Attribute	Remove (deferred)	Grading rules are not described; covered by `outcomeReason`
C-14	Batch	E-07	Conceptual class	Keep as StockLot	The accepted quantity from one line, with batch and dates, is the unit that is allocated and traced
C-15	Harvest information, use-by date	Producers section	Attribute	Convert → StockLot attributes	Optional because producer records vary (A-03)
C-16	Stock	Throughout	Synonym	Merge → StockLot	"Stock" is the set of all stock lots
C-17	Storage (chilled, ambient, temporary)	Receipt section	Attribute	Convert → Product `storageType`	Storage capacity is out of scope; only the required type matters
C-18	Customer	Throughout	Conceptual class	Keep	Places orders and holds preferences

ID	Candidate	Source	Classification	Decision	Rationale
C-19	Household, café, accommodation provider, community programme	Organisation; Orders sections	Attribute	Convert → CustomerType enumeration	Same data and behaviour in the model; differences are only priority, which is unconfirmed (Q-05)
C-20	Pre-paid household box	Orders section	Attribute	Remove (deferred)	Payment is out of scope; treated as a household order
C-21	Order	Throughout	Conceptual class	Keep	Unit of preparation and handover
C-22	Order line	Implied by “several products”	Conceptual class	Keep	Allocation and preferences apply per product
C-23	Substitution preference	E-05	Association	Convert → “permits as substitute” OrderLine–Product	A list of acceptable products, not a thing with its own data
C-24	Reduced quantity, removal	E-06; Orders section	Attribute	Convert → OrderLine Booleans	Simple yes/no customer choices
C-25	Allocation	Orders section	Conceptual class	Keep	Links a quantity of a stock lot to an order line; the basis of traceability (BR-03)
C-26	Shortage / shortfall	Orders section	Attribute (derived)	Convert → `/shortfall` on OrderLine	Requested minus allocated quantity

ID	Candidate	Source	Classification	Decision	Rationale
C-27	Priority rule	Orders section	Business rule	Remove from model	Unconfirmed policy (Q-05); belongs in the uncertainty register
C-28	Label	Orders section	Attribute	Remove	A physical output of packing, not tracked data
C-29	Collection / handover	Collection section	Status	Convert → OrderStatus `HandedOver`	A state of the order
C-30	Missing / damaged / incorrectly allocated goods	Collection section	Conceptual class	Keep as HandoverIssue	Each issue has a type, quantity and decision
C-31	Return	Collection section	Synonym	Merge → HandoverIssue (type Returned)	Same data: quantity and stock decision
C-32	Staff, volunteer, hub manager	Throughout	Role	Convert → HubWorker with `role`	Needed to record who made decisions (NFR-05); roles do not need separate classes
C-33	Driver	Collection section	Role	Remove	Offstage; does not record anything
C-34	Price adjustment, commercial terms	Receipt; Collection sections	Out of scope	Remove	Payment excluded; A-07 records only a note via `outcomeReason`
C-35	Spreadsheet, message, paper	Organisation section	Technology	Remove	Current tools, not domain concepts

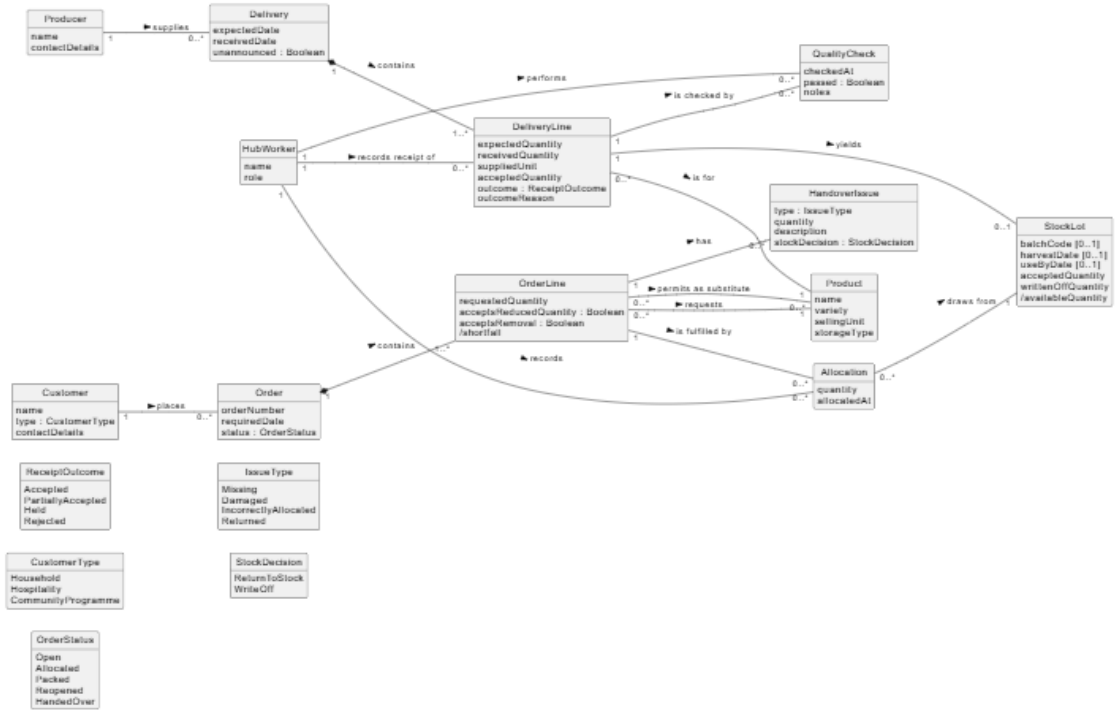
ID	Candidate	Source	Classification	Decision	Rationale
	checklist				
C-36	Route, certification	Collection section	Out of scope	Remove	Explicitly excluded
C-37	Report / producer contributions	Collection section	Derived information	Remove	Derived from existing classes (FR-13); meaning unconfirmed (Q-10)

9.2 Glossary

Term	Definition	Synonyms / notes
Allocation	A quantity from one stock lot assigned to one order line	—
Available quantity	Accepted quantity of a stock lot minus allocated and written-off quantities	Derived; never includes held goods (BR-01)
Delivery	One arrival of goods from a producer, expected or unannounced	“Expected delivery” (merged)
Delivery line	One product within a delivery, with its expected, received and accepted quantities and outcome	—
Held	Received but not yet released by a quality check; not available	—
Handover issue	A problem found at collection (missing, damaged, wrongly allocated) or a returned item, with its stock decision	“Return” (merged)
Order	A customer’s request for several products for a required date	—
Order line	One product within an order, with requested quantity and the customer’s preferences	—
Permitted	A product the customer has	Substitution preference

Term	Definition	Synonyms / notes
substitute	agreed may replace the ordered product	
Producer	A farm or business that supplies goods to the hub	“Supplier” (merged)
Quality check	A recorded inspection of a delivery line that releases or rejects held goods	Inspection
Selling unit	The unit in which the hub sells and allocates a product	Conversion rules open (Q-02)
Shortfall	Requested quantity of an order line minus its allocated quantity	Shortage
Stock lot	The accepted goods from one delivery line, with batch and date details, that can be allocated	Batch, stock

9.3 Domain model



Domain model

Source: `diagrams/domain-model.puml`. 12 conceptual classes, 5 enumerations.

Modelling decisions

1. **DeliveryLine is separate from Delivery.** One delivery can be partly accepted, held and rejected (Receipt section), so the outcome and quantities must belong to each line.
2. **StockLot is separate from DeliveryLine.** A delivery line records what happened at receipt; a stock lot is what can be allocated and traced. Only accepted goods yield a stock lot (multiplicity 0..1), which enforces BR-01 structurally.
3. **Allocation is a class, not a plain association.** It carries its own quantity and time, and one order line can draw from several lots (GH-05). This is what keeps the producer and batch link after stock is split (E-07, BR-03).
4. **Substitution is an association** ("permits as substitute") between OrderLine and Product, because a preference is a list of acceptable products with no data of its own.
5. **Customer types are an enumeration**, not subclasses: the scenario gives no difference in data or behaviour between them, only in priority, which is unconfirmed (Q-05).
6. **Quantities and units stay as attributes.** Modelling a Unit class with conversion factors would invent a policy the hub has not confirmed (Q-02).

Model limitations: priority rules, price adjustments and producer contribution reporting are not represented because their meaning is unresolved (Q-05, Q-08, Q-10).

10. Traceability Matrix

Each row follows one chain from problem evidence to the requirement, its behaviour and the domain concepts that hold its data. Every Must requirement appears at least once.

Evidence	Stakeholder / goal	Requirement	Priority	Rule	Use case	Gherkin / story	Domain concepts
E-02, E-03	SH-01 Hub manager: know what is expected	FR-01	Must	—	UC-01	US-06	Producer, Delivery, DeliveryLine
E-02	SH-01: expected vs accepted are different	FR-02	Must	—	UC-02	GH-01, US-01	DeliveryLine
Receipt section	SH-03 Volunteer: record outcome per line	FR-03	Must	BR-01	UC-02	GH-01, US-01	DeliveryLine, ReceiptOutcome

Evidence	Stakeholder / goal	Requirement	Priority	Rule	Use case	Gherkin / story	Domain concepts
E-08	SH-02 Staff: comparable quantities	FR-04	Should	—	UC-02	—	DeliveryLine (`suppliedUnit`), Product (`sellingUnit`)
E-07	SH-04 Traceability coordinator: batch known	FR-05	Must	BR-03	UC-11	GH-05, US-05	StockLot
E-04	SH-03 Volunteer: received ≠ available	FR-06	Must	BR-01	UC-02, UC-03	GH-02, US-02	DeliveryLine, QualityCheck, StockLot
Orders section	SH-02 Staff: capture orders	FR-07	Must	—	UC-04	GH-03, GH-04	Customer, Order, OrderLine
E-05, E-06	SH-06/07 Customers: get acceptable goods	FR-08	Must	BR-05	UC-04, UC-06	GH-03, GH-04, US-03	OrderLine, Product (permits as substitute)
E-07	SH-04: link survives allocation	FR-09	Must	BR-03	UC-05, UC-07	GH-03, GH-05, US-05	Allocation, StockLot, OrderLine
E-06, E-11	SH-07 Hospitality: early notice	FR-10	Should	—	UC-06	GH-03, US-04	OrderLine (`/shortfall`)
Collection section	SH-02 Staff: resolve handover problems	FR-11	Should	BR-04	UC-08	—	HandoverIssue, StockDecision
E-07	SH-04: trace an item	FR-12	Must	BR-03	UC-09	GH-05, US-05	Allocation, StockLot, Producer
Manager needs	SH-01: see the hub at a glance	FR-13	Should	—	UC-10	US-06	Delivery, StockLot, OrderLine
Visibility	SH-03 Volunteer; SH-	NFR-01	Must	BR-02	UC-02,	—	HubWorker (`role`)

Evidence	Stakeholder / goal	Requirement	Priority	Rule	Use case	Gherkin / story	Domain concepts
statement	01 Manager				UC-07		
E-04	SH-03; SH-06/07	NFR-04	Must	BR-01	UC-03, UC-05	GH-02	StockLot (`/availableQuantity`)
E-07	SH-04; SH-10 Regulator	NFR-05	Must	BR-03	UC-02, UC-03, UC-05	—	HubWorker, QualityCheck, Allocation

Reading the matrix. The strongest chain runs from the traceability coordinator's quote (E-07) through BR-03 to FR-05, FR-09 and FR-12, then to GH-05 and the Allocation and StockLot classes. That chain is why Allocation is a class in the domain model: without it, the requirement could not be satisfied. The weakest links are the Should requirements FR-04 and FR-11, which have no Gherkin scenario yet; both depend on open questions (Q-02, Q-07), so writing scenarios now would mean inventing policy. They would be the first scenarios to add once those questions are answered.

Coverage check: all 9 Must functional requirements and 3 Must non-functional requirements trace to evidence, a use case and at least one domain concept; 8 of the 9 Must FRs also appear in at least one Gherkin scenario (FR-01 is verified by test). No Must requirement is isolated.

11. Reflection

11.1 Important decisions

Separating StockLot from DeliveryLine. The simpler option was to treat a delivery line as the stock itself, with a status field to say whether it was available. I rejected that because of the packing volunteer's point that "received does not mean available" (E-04). A delivery line records what happened at receipt. A stock lot is only what can be allocated. Because a line produces a stock lot only when goods are accepted (0..1), held goods can't be allocated by mistake, and BR-01 is built into the model instead of relying on a status flag being set correctly. I would keep this decision.

Making Allocation a class. A plain association between OrderLine and StockLot would have been simpler. But the traceability coordinator needs to know which producer and batch supplied an item *after* stock has been split (E-07), and one order line can draw from more than one lot. GH-05 shows this, with 36 eggs coming from two batches. An allocation needs its own quantity and time, so it has to be a class. This is the strongest chain in my traceability matrix, from E-07 through BR-03 to FR-12.

Leaving shortage priority as an assumption. I could have written a priority rule, such as hospitality customers first, and turned it into a business rule. The scenario says priority rules are informal, though (E-09), and tells us to identify uncertainty "rather than inventing a confirmed policy". So I recorded it as A-05 (allocate by order date) and Q-05 instead, which keeps the allocation requirements no more certain than the evidence.

11.2 Revisions

My first draft of the scenario interpretation (commit on 12 Oct) was a bullet list of separate problems: quantity differences, units, preferences, priority and traceability. When I rewrote it on 16 Oct, I saw that most of these come from one cause: a quantity moves through several states that are never recorded separately. I rewrote the problem statement around that cause and linked each claim to an evidence ID (E-01 to E-10). I also replaced the single dossier file and the Word document with separate section files so each part could be reviewed on its own.

11.3 Limitations and unresolved questions

Q-05 (shortage priority) limits the analysis most. FR-09, FR-10, GH-03 and risk R-01 all depend on A-05, so it is the first thing I would ask the hub manager. Next is Q-02 (unit conversion), because the availability figures are only as reliable as the conversion between supplied and selling units. FR-04 stays a Should until that is answered. Q-10 means FR-13's "producer contributions" is still undefined.

12. AI Declaration

Tool	Purpose	Assessment areas affected
Claude (Anthropic), via claude.ai	Explaining the brief and scenario; templates and worked examples from a different (library) domain	Sections 1–10
Claude (Anthropic)	Tooling help: how to preview Markdown files and how to work with Gherkin feature files in VS Code	None directly (working environment only)

Representative interactions: Asking for the scenario's core problem and key distinctions; asking for formats and worked examples from a different domain; asking how to view Markdown and use Gherkin in VS Code.

Checks performed: Checked each requirement against the scenario text and quotes; checked that no assumption is presented as a business rule; checked multiplicities against the scenario; rendered both diagrams from PlantUML; checked IDs are consistent across sections; reviewed the drafted reflection against my own reasoning and edited it.

Accepted, modified or rejected: Accepted the general structure and section order from the brief guidance. Modified the library-domain examples, because none of their content applied to the food hub; all requirements, rules, concepts and scenarios were written for Scenario 19.

The full interaction log is in `decisions/ai-use-log.md`.

Appendix A – User Stories

US-01 Record what actually arrived

- **Story:** As a hub volunteer, I want to record the received and accepted quantity for each delivery line, so that the hub knows how much can really be sold.
- **Business value:** Removes the main cause of reallocation errors (E-02).
- **Stakeholders:** SH-01 Hub manager, SH-03 Hub volunteer
- **Links:** FR-02, FR-03, BR-01, UC-02
- **Priority:** Must
- **Acceptance criteria:**
  - AC-01.1 Expected, received and accepted quantities are stored separately.
  - AC-01.2 Only the accepted quantity becomes available.
  - AC-01.3 A partially accepted line records the remainder as held or rejected.
- **Gherkin:** GH-01
- **Questions / assumptions:** Q-02, A-02

US-02 Keep unchecked goods out of stock

- **Story:** As a packing volunteer, I want held goods to be excluded from available stock, so that I never pack goods that have not passed checking.
- **Business value:** Prevents unchecked produce reaching customers (E-04).
- **Stakeholders:** SH-03 Hub volunteer, SH-06/07 Customers
- **Links:** FR-06, BR-01, NFR-04, UC-02, UC-03
- **Priority:** Must
- **Acceptance criteria:**
  - AC-02.1 A held line adds 0 to available stock.

    - AC-02.2 Held quantity becomes available only after a recorded quality release.
- **Gherkin:** GH-02
- **Questions / assumptions:** Q-07, A-08

US-03 Respect substitution preferences

- **Story:** As a customer, I want my acceptable substitutes and reduced-quantity preference to be recorded, so that I am not sent alternatives I would refuse.
- **Business value:** Fewer rejected items and complaints at handover (E-05).
- **Stakeholders:** SH-06 Household customer, SH-02 Hub staff member
- **Links:** FR-08, BR-05, UC-04, UC-06
- **Priority:** Must
- **Acceptance criteria:**
    - AC-03.1 Each order line can list zero or more permitted substitutes.
    - AC-03.2 A substitute that is not permitted cannot be allocated.
- **Gherkin:** GH-04
- **Questions / assumptions:** Q-06, A-06

US-04 Warn hospitality customers early

- **Story:** As a hospitality customer, I want to be told about a reduced quantity before delivery, so that I can plan my menu.
- **Business value:** Keeps regular hospitality customers (E-06).
- **Stakeholders:** SH-07 Hospitality customer, SH-02 Hub staff member
- **Links:** FR-10, UC-06
- **Priority:** Should
- **Acceptance criteria:**
    - AC-04.1 Any order line allocated below its requested quantity shows the shortfall.
    - AC-04.2 The customer is flagged for contact before the delivery date.
- **Gherkin:** GH-03
- **Questions / assumptions:** Q-05, A-05

US-05 Trace an item to its source

- **Story:** As the traceability coordinator, I want to see the producer and batch of any allocated item, so that I can respond to a quality or safety query.
- **Business value:** Restores traceability lost when stock is split (E-07).
- **Stakeholders:** SH-04 Traceability coordinator, SH-10 Regulator
- **Links:** FR-05, FR-09, FR-12, BR-03, NFR-05, UC-09
- **Priority:** Must

- **Acceptance criteria:**
  - AC-05.1 Every allocation is linked to exactly one stock lot.
  - AC-05.2 A trace lists each producer and batch contributing to the order line, with quantities.
- **Gherkin:** GH-05
- **Questions / assumptions:** Q-03, A-03

US-06 See the hub at a glance

- **Story:** As the hub manager, I want a summary of expected vs accepted deliveries, held stock and shortfalls, so that I can act before orders are packed.
- **Business value:** Replaces manual reporting from spreadsheets (E-01).
- **Stakeholders:** SH-01 Hub manager
- **Links:** FR-13, UC-10
- **Priority:** Should
- **Acceptance criteria:**
  - AC-06.1 The summary shows expected vs accepted quantity per producer.
  - AC-06.2 The summary shows held stock, available stock, shortfalls and substitutions.
- **Gherkin:** none (verified by inspection)
- **Questions / assumptions:** Q-10